UNIVERSITÀ DEGLI STUDI FIRENZE

Montecarlo Simulations for CI Actual Coverage

Wald & Wilson

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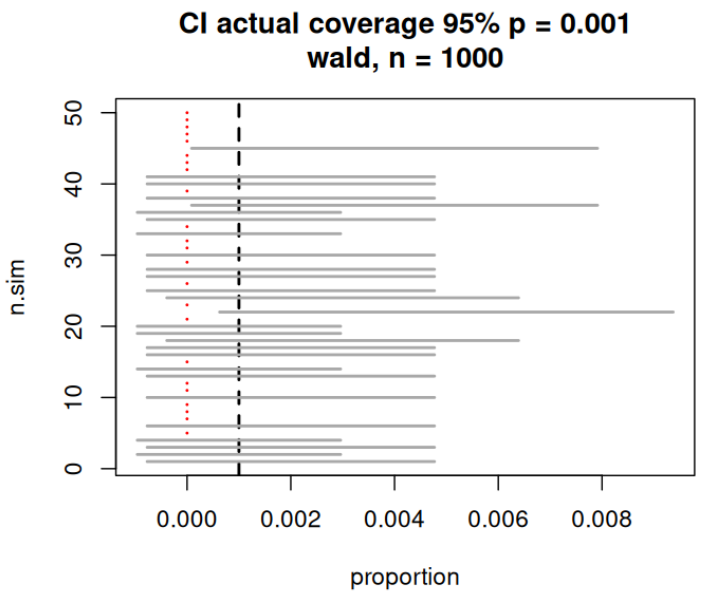
CI segment plots

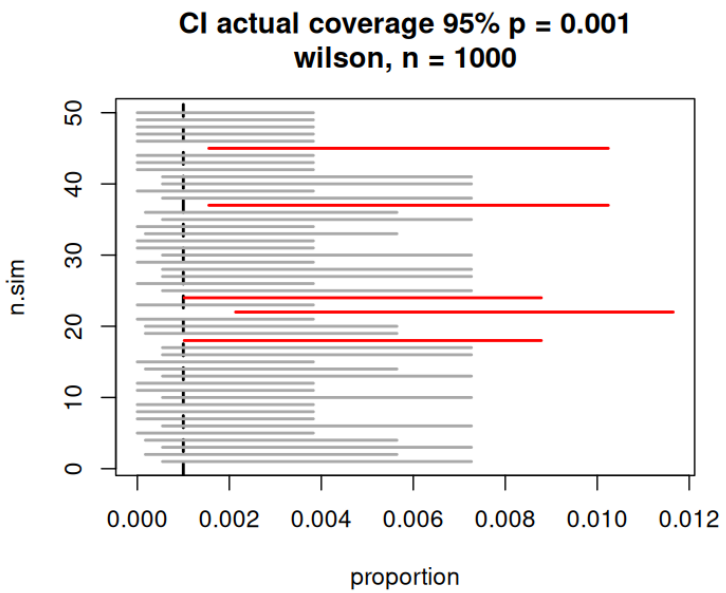
Wald CI:

$$CI_{Wald} = \hat{p} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Wilson CI:

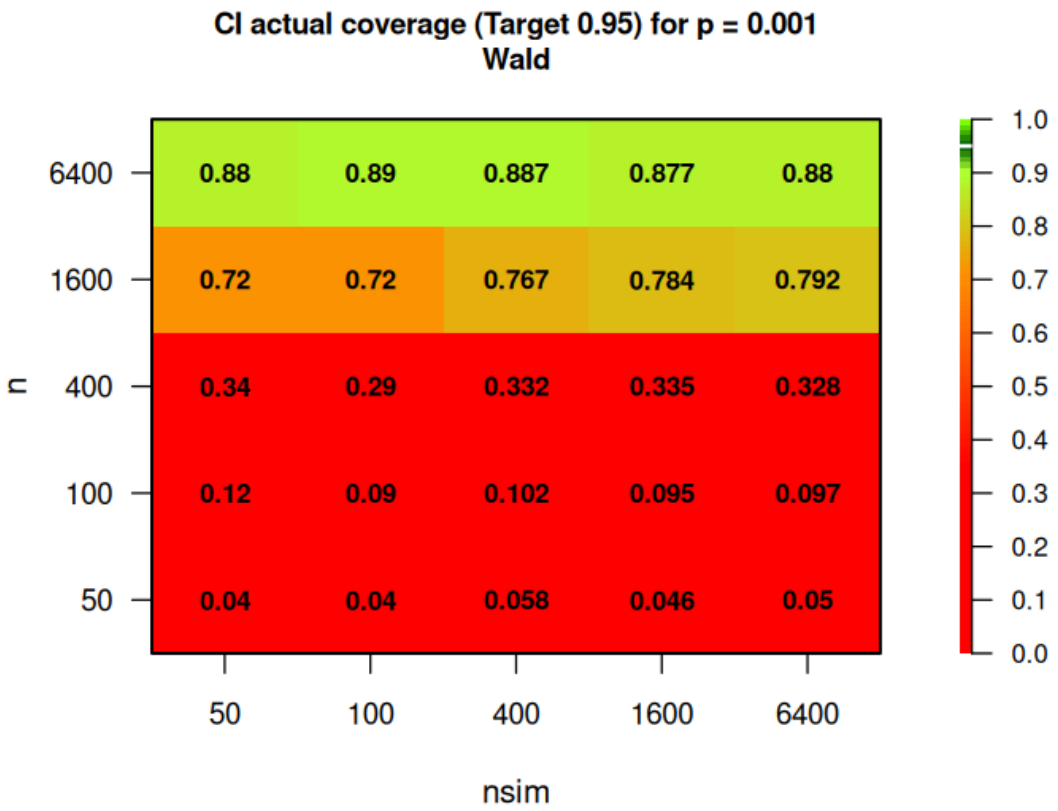
$$CI_{Wilson} = \frac{1}{1+\frac{z^2}{n}}\left(\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}\right)$$

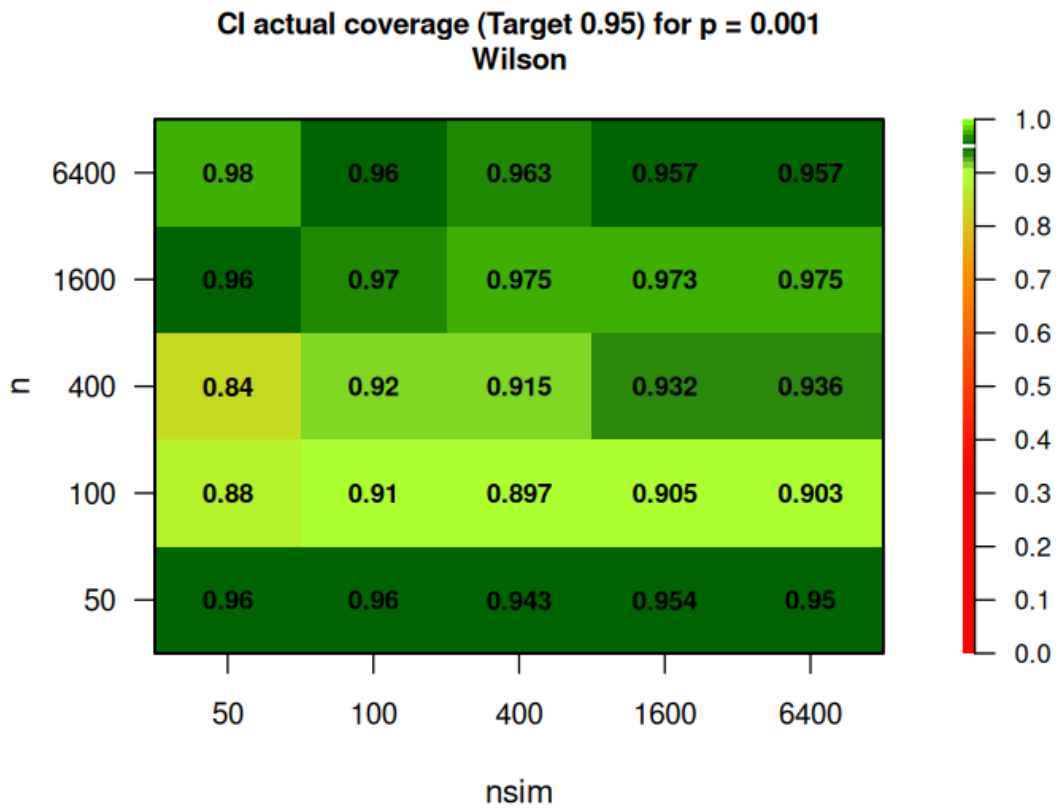




- Wald CIs fail for very small values of p
- Wilson CIs always falls in [0,1]

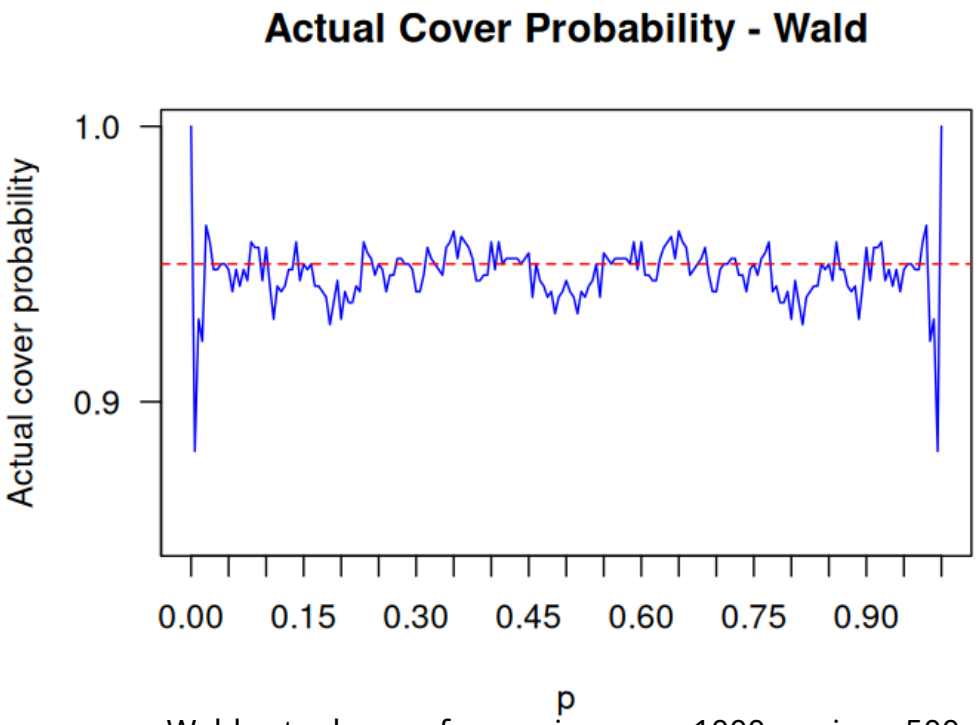
Heatmap Plot



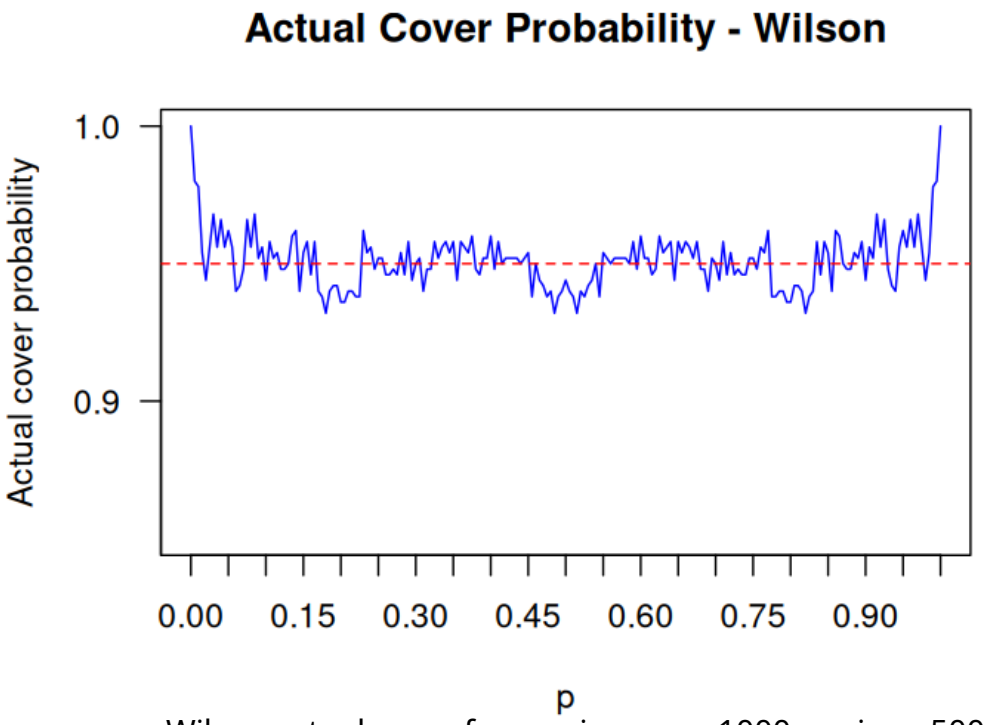


- CI heatmap for wald. The best results are obtained with a big number of samplings.
- Wilson test manages to obtain better actual coverage with small values of p and n.

Actual Coverage Plot



Wald actual cover for varying p n = 1000, n_sim = 500.



Wilson actual cover for varying p n = 1000, n_sim = 500.

- Wald has a worst actual cover probability for values of p closed to the edge compared to Wilson.
- For values of p far from the edges Wilson and Wald obtain the same actual cover probability.

Conclusions

Overall Wilson performs better than Wald, especially for values of p near the edges of the domain.

References:

Alessandra, Mattei (2025). Slides05_HyphotesisTests [Slide di PowerPoint].
Alessandra, Mattei (2025). Slides06_CI [Slide di PowerPoint].
Anna, Gottard (2026). slides01_FSL [Slide di PowerPoint].
Anna, Gottard (2026). Lab01 [File Html].
Claude: https://claude.com
Gemini: https://gemini.google.com